

the integration, our chip measures current, voltage, and temperature across multiple battery cells, performs active cell balancing, and includes a Tru-EIS engine for continuous health diagnostics.

The chip also incorporates a lockstep microcontroller for automotive safety, an AI accelerator for advanced battery estimation algorithms, integrated power management, and gate drivers that can safely disconnect the battery during fault conditions.

Can you shed some light on the 'EIS Engine'?

A key highlight of our chip is the integration of electrochemical impedance spectroscopy (EIS) directly into the chip, enabling continuous, real-time monitoring of battery health, similar to an ECG.

The system continuously monitors electrochemical reactions within each cell, enabling accurate measurement of degradation and ageing rather than relying on predictive modelling. This significantly improves battery longevity, reliability, and performance.

Traditional battery management systems assess battery health only periodically. Our approach enables continuous 24/7 monitoring of every battery cell using EIS integrated directly into the chip. This allows real-time insight into battery ageing, early fault detection, improved safety, and more efficient energy usage.

What role does AI play inside a battery management system?

Batteries generate large amounts of operational data, and their behaviour changes as they age. AI enables the BMS to adapt dynamically by learning from historical usage patterns.

Advanced models can predict ageing behaviour, optimise charging strategies, and improve performance over time. The onboard AI accelerator ensures the chip remains

future-ready as battery algorithms continue evolving.

How does a single-chip BMS compare with existing solutions in terms of reliability and competition?

Traditional BMS architectures distribute functions across several chips, creating communication overhead and multiple physical failure points. Our single-chip design removes most inter-chip connections, allowing faster data access, higher accuracy, and improved reliability.

While established players like Texas Instruments, NXP Semiconductors, and Qorvo offer advanced BMS solutions, our differentiation lies in integrating all core BMS functions, along with Tru-EIS capability, into a single tightly integrated IC.

How does your IC minimise power loss and manage energy consumption?

Power efficiency comes mainly from integration. Traditional BMS designs consume energy through constant communication among multiple chips, whereas our single-chip architecture eliminates much of this overhead by sharing internal memory.

The IC also operates across multiple power states, dynamically adjusting energy consumption based on vehicle conditions to achieve very low standby power while maintaining readiness.

How do you validate performance, support different battery chemistries, and technically differentiate your chip from existing solutions?

Validation involves extensive testing across voltage ranges, temperatures, and automotive operating conditions using dedicated evaluation hardware.

The chip delivers high measurement accuracy for current, voltage, and battery health monitoring while remaining chemistry-agnostic.

A key differentiator is the

integrated hardware abstraction layer, which simplifies software development by handling low-level hardware control internally.

This enables manufacturers to focus on battery algorithms and optimisation rather than complex hardware integration, thereby supporting a more software-defined BMS platform.

What are the key technical challenges you are facing, and how are you addressing them?

Designing a single-chip BMS is far more complex than building a conventional IC because it combines multiple engineering disciplines: digital design, precision analogue and mixed-signal electronics, high-voltage operation, automotive functional safety, and deep battery system expertise. Automotive environments demand redundancy and fail-safe operation, meaning the chip must continue functioning safely even if parts of the system fail.

Our team's prior experience across automotive, analogue, digital, and battery IC development enables us to integrate these domains into a single architecture while maintaining safety and reliability.

How will this innovation impact electric vehicle costs and battery lifespan?

Higher integration can reduce overall electronics cost, which may help lower EV system prices over time.

More importantly, continuous and accurate battery monitoring improves safety and significantly extends battery lifespan.

Predictive diagnostics can help users avoid premature battery replacement, lowering total cost of ownership even if overall vehicle prices remain unchanged. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/augint1>

